

Performance & Model Validation Testing

1. Model Validation (ML Performance)

The core Logistic Regression classifier was validated using an 80/20 train-test split on 2,200 dataset records. Outliers in the phosphorous parameter were cleaned using Interquartile Range (IQR) filtering.

Validation Performance Summary:

- **Overall Model Accuracy:** 94.09%
- **Precision (Average):** 94.20%
- **Recall (Average):** 94.09%
- **F1-Score (Average):** 94.11%

Classification Report snippet:

- Rice: Precision 1.00, Recall 0.95
- Maize: Precision 0.89, Recall 0.93
- Chickpea: Precision 1.00, Recall 1.00
- Pomegranate: Precision 1.00, Recall 0.98

2. Application Performance & Load Testing

The local Flask application was tested under multiple loads:

- **Average Page Load Time:** < 80ms (Home screen, About, and Login).
- **Prediction Request Processing Time:** < 120ms (Includes database write transactions and ML inference).
- **Concurrent Users Handling:** Verified stable handling of up to 50 concurrent simulation requests locally without database thread locking.